DS	**Session**	Tree 1	**Question Information**	Level 1 Challenge 65

Question description

Given an array of n integers, your task is to process q queries of the form: what is the minimum value in range [a,b]?

Constraints

- 1≤n,q≤2·10^5
- 1≤xi≤10^9
- 1≤a≤b≤n

Input

The first input line has two integers n and q: the number of values and queries.
The second line has n integers x1,x2,…,xn: the array values.
Finally, there are q lines describing the queries. Each line has two integers a and b: what is the minimum value in range [a,b]?

Output
Print the result of each query.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
8 4
3 2 4 5 1 1 5 3
2 4
5 6
1 8
3 3
```

EXPECTED OUTPUT

```
2
1
1
4
```

Test Case 2

INPUT (STDIN)

```
10 4
13 12 14 51 11 1 5 3 4 6
3 4
3 6
4 8
1 3
```

EXPECTED OUTPUT

```
14
1
1
12
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void build(int *aa,int k,int l,int r)
```

Test Case 2

KEYWORD

```
int query(int k,int l,int r,int ql,int qr)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

390

Test Case 3

NLOC

40

Code

solution.c

```c
#include <stdio.h>

#define INF 1000000009
int tree[800005];

int minVal(int a, int b) { return (a < b) ? a : b; }

void build(int *aa, int k, int l, int r) {
    if (l == r) { tree[k] = aa[l]; return; }
    int mid = (l + r) / 2;
    build(aa, 2 * k, l, mid);
    build(aa, 2 * k + 1, mid + 1, r);
    tree[k] = minVal(tree[2 * k], tree[2 * k + 1]);
}

int query(int k, int l, int r, int ql, int qr) {
    if (ql <= l && r <= qr) return tree[k];
    if (r < ql || l > qr) return INF;
    int mid = (l + r) / 2;
    return minVal(query(2 * k, l, mid, ql, qr), query(2 * k + 1, mid + 1, r, ql, qr));
}

int main() {
    int n, q;
    if (scanf("%d %d", &n, &q) != 2) return 0;
    int a[n + 1];
    for (int i = 1; i <= n; i++) scanf("%d", &a[i]);
    build(a, 1, 1, n);
    while (q--) {
        int l, r;
        scanf("%d %d", &l, &r);
        printf("%d\n", query(1, 1, n, l, r));
    }
    return 0;
}
```

Output

```
Problems  Output  Debug Console  Terminal  Ports        powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
8 4
3 2 4 5 1 1 5 3
2 4
5 6
1 8
3 3
2
1
1
4
PS C:\C Programming>
```